Current model (ordinary-matter shell + CoE core) vs. A_F, term by term

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1. Model and where each term comes from. Sedov–Taylor flow $\vec{V} = k\,\vec{x}$ (position from the CoE; Taylor 1950; Sedov 1959), decelerated by a point core M_c at the CoE and a uniform shell Ω_s between $R_{\rm in}$ and $R_{\rm out}$ (fixed in space; Earth and all sources inside it). Spherical symmetry makes the field radial, $\vec{V} = h(q)\,\vec{q}$, with $q = s/D$ (D = CoE–Earth distance). Linear perturbation of the code's integration (from $\tau_i = 0.01$, $Ht \simeq 1$):

$$\frac{h(q)}{k} \simeq 1 - \tfrac{1}{2}\Big[\underbrace{\tfrac{M_c}{M_{\rm crit}(D)}\big(\tfrac{1}{\tau_i}-1\big)q^{-3}}_{\text{core tidal term}} + \Omega_s\Big(\underbrace{\tfrac{1}{2}}_{\text{uniform: } m=0 \text{ only}} \underbrace{-\tfrac{3}{2}r_{\rm in}^2 q^{-2} + r_{\rm in}^3 q^{-3}}_{\text{hole } (r_{\rm in}=R_{\rm in}/D)}\Big)\Big], \qquad \beta \equiv \frac{h'}{h}\Big|_{q=1}, \quad \gamma \equiv \frac{h''}{h}\Big|_{q=1}.$$

Chris's A: $a_f = |\Delta\vec{V}|$/separation with sources at $R = 14.3$ Gpc, $\varepsilon \equiv R/D$, $\mu = \cos b_0 \cos\psi$, $\psi = \ell - \ell_0$ (ℓ_0 = anti-CoE direction). Exactly $a_f \propto |h(q_s)(\hat{x} + \varepsilon\hat{n}) - h(1)\hat{x}|/\varepsilon$; to first order in ε (the higher orders are the r/D terms):

$$a_f^2 \propto 1 + B\mu^2 + \varepsilon\,(P\mu + Q\mu^3), \qquad B = 2\beta + \beta^2, \quad P = \beta(3+\beta), \quad Q = (1+\beta)(\gamma+\beta) - 2\beta.$$

Take the square root, then use $\cos^2\psi = \frac{1}{2}(1+\cos 2\psi)$, $\cos^3\psi = \frac{1}{4}(3\cos\psi + \cos 3\psi)$, $\cos^4\psi = \frac{3}{8} + \frac{1}{2}\cos 2\psi + \frac{1}{8}\cos 4\psi$, $\cos^5\psi = \frac{1}{16}(10\cos\psi + 5\cos 3\psi + \cos 5\psi)$ ($b_0 = 0$ as in the code):

$$\frac{a_f}{\langle a_f\rangle} = 1 + a_1\cos\psi + a_2\cos 2\psi + a_3\cos 3\psi + \ldots, \quad a_2 = \frac{\frac{B}{4} - \frac{B^2}{16}}{m_0}, \quad a_1 = \frac{\varepsilon}{2m_0}\Big[P + \tfrac{3}{4}\big(Q - \tfrac{PB}{2}\big) - \tfrac{5QB}{16}\Big], \quad a_3 = \frac{\varepsilon}{2m_0}\Big[\tfrac{1}{4}\big(Q - \tfrac{PB}{2}\big) - \tfrac{5QB}{32}\Big],$$

with $m_0 = 1 + \frac{B}{4} - \frac{3B^2}{64}$. For a pure point mass ($\gamma = -4\beta$, $\beta \to 3K$) this is the earlier line-of-sight form $K[(3\mu^2-1) + \varepsilon(\frac{9}{2}\mu - \frac{15}{2}\mu^3)]$. The geometry gives only $\cos m\psi$ terms: even harmonics set by β, odd ones by εQ. *Rotation* (the sine factor, $A = a_f[1 - {\rm Amp}\sin(\psi+\varphi)]$) adds $C_1 = a_1 - {\rm Amp}\sin\varphi(1 + \frac{a_2}{2})$, $S_1 = -{\rm Amp}\cos\varphi(1 - \frac{a_2}{2})$, $C_3 = a_3 - \frac{{\rm Amp}\,a_2}{2}\sin\varphi$, $S_3 = -\frac{{\rm Amp}\,a_2}{2}\cos\varphi$ (C_m, S_m: $\cos m\psi, \sin m\psi$ parts). **Latitude:** the code evaluates the model in the Galactic plane only (one direction per degree, no $|b|$ cut, no sector average), so the match is done the same way. A tilt b_0 would give $B \to B\cos^2 b_0$ and ε-terms $\propto \cos b_0, \cos^3 b_0$, degenerate with β and εQ; therefore $b_0 = 0$.

2. A_F rotated onto the model axis ($\ell_0 = 14.96°$, the 2ℓ peak), $c_m\cos m\ell + s_m\sin m\ell = C_m\cos m\psi + S_m\sin m\psi$:

	A_0	A_1, ℓ_1	A_2, ℓ_2	A_3, ℓ_3	C_1	S_1	C_3	S_3
A_F	0.9965	0.0525, 55.7°	0.0734, 14.96°	0.0320, 67.0°	0.0398	0.0343	−0.0292	0.0130

3. Term-by-term solution (Newtonian, series above, point-core shape $\gamma = -4\beta$; exact least-squares in brackets):

Term	A_F coefficient	Analytic expression	Solved value
2ℓ phase	$\ell_2 = 14.96°$	ℓ_0 (axis; $\beta > 0$: fast along it)	$\ell_0 = 14.96°$ (12.8°)
2ℓ amplitude	$C_2 = 0.0734$	$a_2(\beta)$	$\beta = 0.161$ (0.210)
1ℓ (rotation)	$C_1 = 0.0398$, $S_1 = 0.0343$	C_1, S_1 of Sec. 1	Amp $= -0.068$, $\varphi = 1.02$ (-0.072, 1.03)
3ℓ cos	$C_3 = -0.0292$	$a_3(\varepsilon Q) - \frac{{\rm Amp}\,a_2}{2}\sin\varphi$	$\varepsilon Q = -0.308 \Rightarrow \varepsilon = 0.348$, $D = 41$ Gpc (47)
3ℓ sin	$S_3 = 0.0130$	$-\frac{{\rm Amp}\,a_2}{2}\cos\varphi$ (check)	predicted 0.0013: **not produced**
$m = 0$	$A_0 = 0.9965$	$N[1 - \frac{1}{2}{\rm Amp}\,a_1\sin\varphi]$	$N = 0.9971$
mass	–	$\beta \simeq \frac{3K}{1-K}$, $K = \frac{1}{2}\frac{M_c}{M_{\rm crit}(D)}(\frac{1}{\tau_i} - 1)$	$M_c/M_{\rm crit}(D) \approx 0.0010$ (linear; ≈0.0017 nonlinear)

The 3ℓ step fixes only εQ: with the 0.909-fit shape ($\gamma/\beta = -14.5$) it gives $D = 114$ Gpc (exact 135), with the $\Omega_s = 0.049$ shape ($\gamma/\beta = -3.7$) $D = 39$ Gpc (exact 45). Every Newtonian solution reaches $R^2 = 0.987$ vs A_F (0.905 vs Migkas).

4. Cross-check: analytic series vs exact numeric projection of Chris's code (geometric part, a_m of $a_f/\langle a_f\rangle$; the code's own β, γ from $h(q)$):

Fit	D [Gpc]	ε	β ($\beta_{\rm lin}$)	γ	a_1 series / exact	a_2 series / exact	a_3 series / exact	a_4 exact
Newt. Ω_s free	93.7	0.153	0.181 (0.26)	−2.626	−0.1082 / −0.0844	0.0813 / 0.0807	−0.0448 / −0.0347	0.0027
Newt. Ω_s=0.049	49.2	0.291	0.157 (0.26)	−0.575	−0.0110 / −0.0126	0.0717 / 0.0764	−0.0240 / −0.0256	0.0069
Relativistic	32.9	0.036[a]	0.186[b]	–	0 / −0.0043	0.0849 / 0.0768	0 / −0.0176	0.0021

[a] r/D, $r = 1.19$ Gpc. [b] $H_r/H_\perp - 1$ at Earth.

The linear formula for $h(q)$ overestimates β by about 1.5–1.7 (the core pull at $\tau \sim \tau_i$ is not small), so β, γ are read from the code; with them the series agrees with the exact projection to $\lesssim$30% (a_2 to 6%). Using the exact radial field $h(q)$ reproduces the code to 10^{-4}. For the relativistic fit $A \propto D_L^{-1}$ at $z = 0.3$ (sampling $r = 1.19$ Gpc); the local shear gives $a_2 \simeq \frac{\beta/2}{1+\beta/2}$ with $\beta = H_r/H_\perp - 1$, and $r/D = 0.036$ would make $a_1, a_3 \approx 0$, but the exact LTB curve has $a_3 = -0.018$ (light-cone/shell-edge structure, not r/D).

5. The three numerical fits, harmonic by harmonic (full model with sine; amplitude @ peak longitude):

	R^2 Migkas	R^2 vs A_F	A_1	A_2	A_3	$M_c/M_{\rm crit}(D)$, D
A_F	–	1	0.0525 @ 56°	0.0734 @ 15°	0.0320 @ 67°	–
Newtonian, Ω_s free	0.909	0.987	0.0527 @ 56°	0.0737 @ 12°	0.0298 @ 72°	0.0027, 93.7 Gpc
Newtonian, Ω_s=0.049	0.893	0.977	0.0530 @ 55°	0.0754 @ 13°	0.0236 @ 72°	0.0017, 49.2 Gpc
Relativistic, all free	0.885	0.963	0.0528 @ 56°	0.0763 @ 14°	0.0160 @ 72°	0.083, 32.9 Gpc

All three get A_1 (via the rotation) and A_2 (via β) right; they differ only in A_3, which ranks them (0.909 > 0.893 > 0.885). The hand solution ($\beta \approx 0.16$, Amp ≈ -0.07, $\varphi \approx 1.0$, $\ell_0 \approx$ 13–15°, $\varepsilon Q \approx -0.31$) matches the Ω_s=0.049 fit ($\beta = 0.157$, Amp $= -0.062$, $\varphi = 0.93$, $\ell_0 = 13°$, $D = 49$ vs 41–45 Gpc).

6. Degeneracies and what the geometry cannot do. (i) *Only β is fixed by 2ℓ.* β depends on $M_c/M_{\rm crit}(D)$, Ω_s and $R_{\rm in}/D$, all ratios: the core's tidal field in units of H^2 is $GM_c/(H^2D^3) = \frac{1}{2}M_c/M_{\rm crit}(D)$. D enters only through $\varepsilon = r/D$ in the odd terms. (ii) *Newtonian:* $r = R = 14.3$ Gpc, so 3ℓ fixes εQ. D trades against the curvature γ of $h(q)$, i.e. the shell structure ($D \approx$ 41–47, 39–45 and 114–135 Gpc for the point-core, Ω_s=0.049 and 0.909 shapes, same R^2). (iii) *Relativistic:* $r = 1.19$ Gpc, so $r/D \le 0.036$ and the odd terms do not depend on D. Along the fitted valley $a_2 = 0.076$–0.078 and $a_3 = -0.016$–-0.018 stay the same for $D = 33$–7×10^6 Gpc while $M_c/M_{\rm crit}(D) = 0.061$–$0.083$. Hence the LTB fit pins $M_c/M_{\rm crit}(D)$, not D. (iv) *Not produced by the geometry:* the dipole is along the axis ($a_1\cos\psi$, peak at ℓ_0 or $\ell_0 + 180°$), but A_F peaks at $\ell_1 = 55.7°$, so the off-axis dipole needs the rotation (sine). The 3ℓ phase offset ($S_3 = +0.013$ vs ≤ 0.002 from rotation) cannot be made: the model's 3ℓ peak is locked to $\ell_0 + 60°$ (data $\ell_3 = 67°$ vs $\ell_0 + 60 \approx$ 73–75°). This missing "3ℓ transform" is the main residual of every fit.

7. Input parameters implied by the term-by-term solution vs. Table 1 of universe_bestfit_thinshell.pdf. Each column keeps that fit's shell assumption. Newtonian Ω_s free: $\Omega_s = 1.012$, thin shell $R_{\rm in} = D - R$, $R_{\rm out} = D + R$. Newtonian Ω_s=0.049: $R_{\rm in} = D - 1.028\,R$, $R_{\rm out} = D + 1.475\,R$. Relativistic: $R_{\rm in} = D - 1.040\,R$, $R_{\rm out} = D + 1.171\,R$. The unknowns are solved from the Sec. 3 equations (2ℓ phase $\to \ell_0$, 2ℓ amplitude $\to \beta \to M_c/M_{\rm crit}$, $1\ell \to$ Amp, φ, 3ℓ cos $\to \varepsilon Q \to D$; C_m, S_m, m = 1–3, by least squares). The $a_m(\beta, \gamma)$ come from the code's exact $h(q)$, not the linear formula. Masses use $H_0 = 70$, $\rho_s = \Omega_s\rho_c$.

	Newtonian, Ω_s free		Newtonian, Ω_s=0.049		Relativistic, all free	
Quantity	term-by-term	Table 1	term-by-term	Table 1	term-by-term (Newt., r=1.19)	Table 1 (LTB)
D/R	6.474	6.554	2.589	3.439	not fixed	2.303
D	2.857×10^{24} km (92.6 Gpc)	2.892×10^{24} km (93.7 Gpc)	1.143×10^{24} km (37.0 Gpc)	1.517×10^{24} km (49.2 Gpc)	any D (tried 33–330)	1.016×10^{24} km (32.9 Gpc)
$R_{\rm in}$	78.3 Gpc	79.4 Gpc	22.3 Gpc	34.5 Gpc	$D - 14.9$ Gpc	18.1 Gpc
$R_{\rm out}$	106.9 Gpc	108.0 Gpc	58.1 Gpc	70.3 Gpc	$D + 16.7$ Gpc	49.7 Gpc
M_c	2.38×10^{54} kg 1.20×10^{24} $M_\odot$	2.47×10^{54} kg 1.24×10^{24} $M_\odot$	9.41×10^{52} kg 4.73×10^{22} $M_\odot$	2.31×10^{53} kg 1.16×10^{23} $M_\odot$	$\propto D^3$ (7.1×10^{52} kg at 32.9 Gpc)	3.38×10^{54} kg 1.70×10^{24} $M_\odot$
$M_c/M_{\rm crit}(D)$	0.00265	0.00265	0.00164	0.00171	**0.0016–0.0018**[a]	0.0835
$M(<D)$, total	3.62×10^{56} kg	3.72×10^{56} kg	2.29×10^{54} kg	4.56×10^{54} kg	$\propto D^3$	5.04×10^{54} kg
$M_{\rm shell}$, total	8.50×10^{56} kg	8.71×10^{56} kg	1.03×10^{55} kg	1.70×10^{55} kg	$\propto D^2$	6.48×10^{54} kg
M within R	3.35×10^{54} kg	3.35×10^{54} kg	1.62×10^{53} kg	1.62×10^{53} kg	1.62×10^{53} kg	1.62×10^{53} kg
Ω_s	1.012[b]	1.012	0.049[b]	0.049	0.049[b]	0.049
ℓ_0	13.2°	12.8°	12.8°	13.0°	15.0°	13.5°
Amp	−0.1246	−0.1240	−0.0637	−0.0622	−0.0528	−0.0558
φ	1.267 rad	1.264 rad	0.941 rad	0.929 rad	0.819 rad	0.848 rad
R^2 vs Migkas	0.909	0.909	0.881	0.893	0.824[c]	0.885
R^2 vs A_F	0.987	0.987	0.970	0.977	0.899[c]	0.963

[a] Newtonian code, $\tau_i = 0.01$, sampling $r = 1.19$ Gpc: the r/D-odd terms vanish, so D is not fixed; any D works with $M_c/M_{\rm crit}(D) \approx 0.0017$ (0.0018, 0.0017, 0.0016 at D = 33, 100, 330 Gpc). The exact LTB curve instead gives $M_c/M_{\rm crit}(D) = 0.061$–$0.083$ for any D (Sec. 6). [b] Held fixed (that column's shell assumption). [c] At $D = 32.9$ Gpc. The harmonic solve weights C_m, S_m equally, not the 71 points, so its R^2 can sit slightly below the point fit.

The **0.909 fit** is recovered: D 92.6 vs 93.7 Gpc, the same $M_c/M_{\rm crit}$, and ℓ_0, Amp, φ to 3%. For **Ω_s=0.049** the harmonic solve gives the same $M_c/M_{\rm crit}$ (0.00164 vs 0.00171), ℓ_0, Amp and φ, but a smaller D (37 vs 49 Gpc): the harmonic weighting pulls toward a stronger 3ℓ term. **τ_i test** (Chris's hypothesis that the code pins $M_c/M_{\rm crit}\cdot(1/\tau_i - 1)$). The Ω_s=0.049 fit was redone at other start times, with D, M_c, both shell edges, ℓ_0, Amp and φ free:

τ_i	$M_c/M_{\rm crit}(D)$	$M_c/M_{\rm crit}\cdot(1/\tau_i - 1)$	D [Gpc]	R^2
0.1	0.0112	0.101	55.4	0.894
0.01 (Table 1)	0.0017	0.169	48.7	0.893
0.001	0.0015	1.491	47.0	0.892

The hypothesis holds only partly. From $\tau_i = 0.01$ to 0.1, $M_c/M_{\rm crit}$ rises 6.5× (11× predicted). From 0.01 to 0.001 it falls only 13% (10× predicted), because the core's own size ($L_c = 0.005\,D$) cuts off the early pull. So the Newtonian $M_c/M_{\rm crit}$ is set by the code's start time and core size, and $M_c/M_{\rm crit}\cdot(1/\tau_i - 1)$ is not constant (0.10–1.5). The ∼50× gap to the relativistic 0.083 is mostly this regularisation: at $\tau_i = 0.1$ it shrinks to 7×. The LTB value (single bang time, no τ_i) is the physically calibrated one.

References. Taylor, G. I. 1950, Proc. R. Soc. Lond. A, 201, 159. Sedov, L. I. 1959, Similarity and Dimensional Methods in Mechanics (Academic Press). Migkas, K., et al. 2020, A&A, 636, A15. Scripts: A_equals_fourier/thinshell/tm_core.py, termmatch.py, relvalley.py, make_tex.py, solve_inputs.py, taui_test.py; numbers in termmatch.json, solve_inputs.json, taui_test.json.